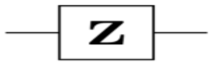
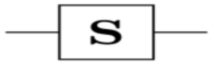
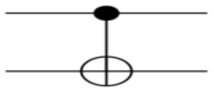
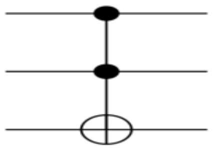
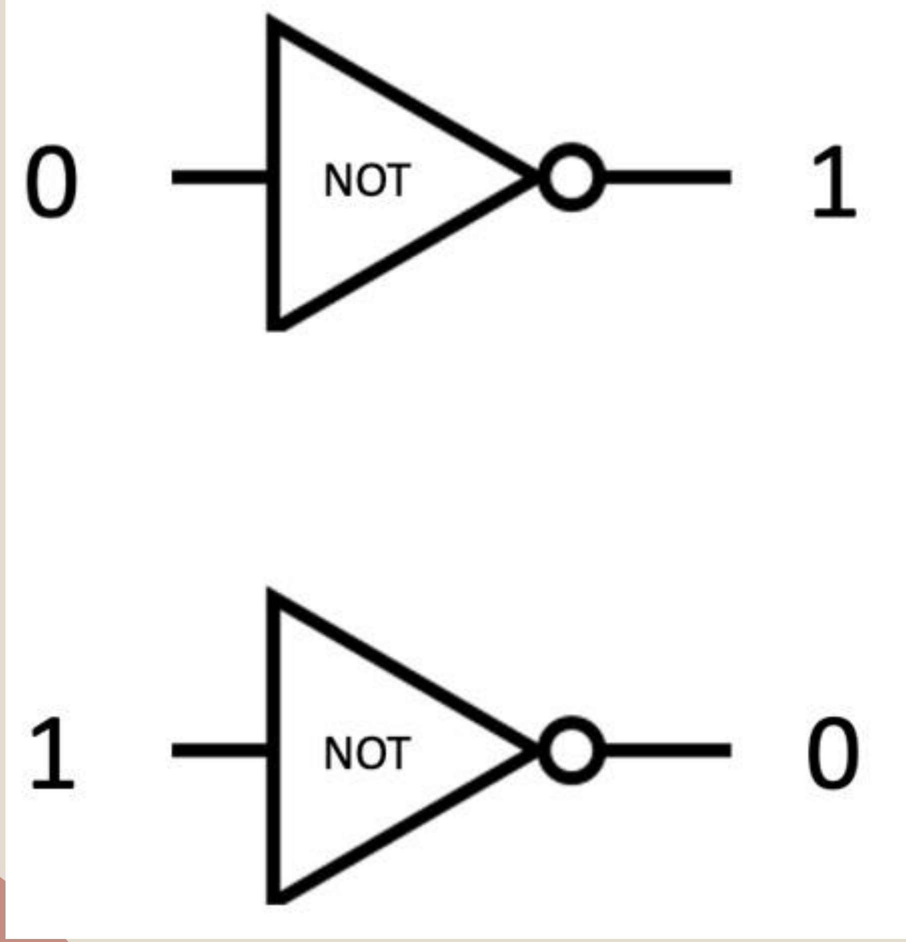
Quantum Gates & Operators

Foreword: The importance & nature of quantum gates

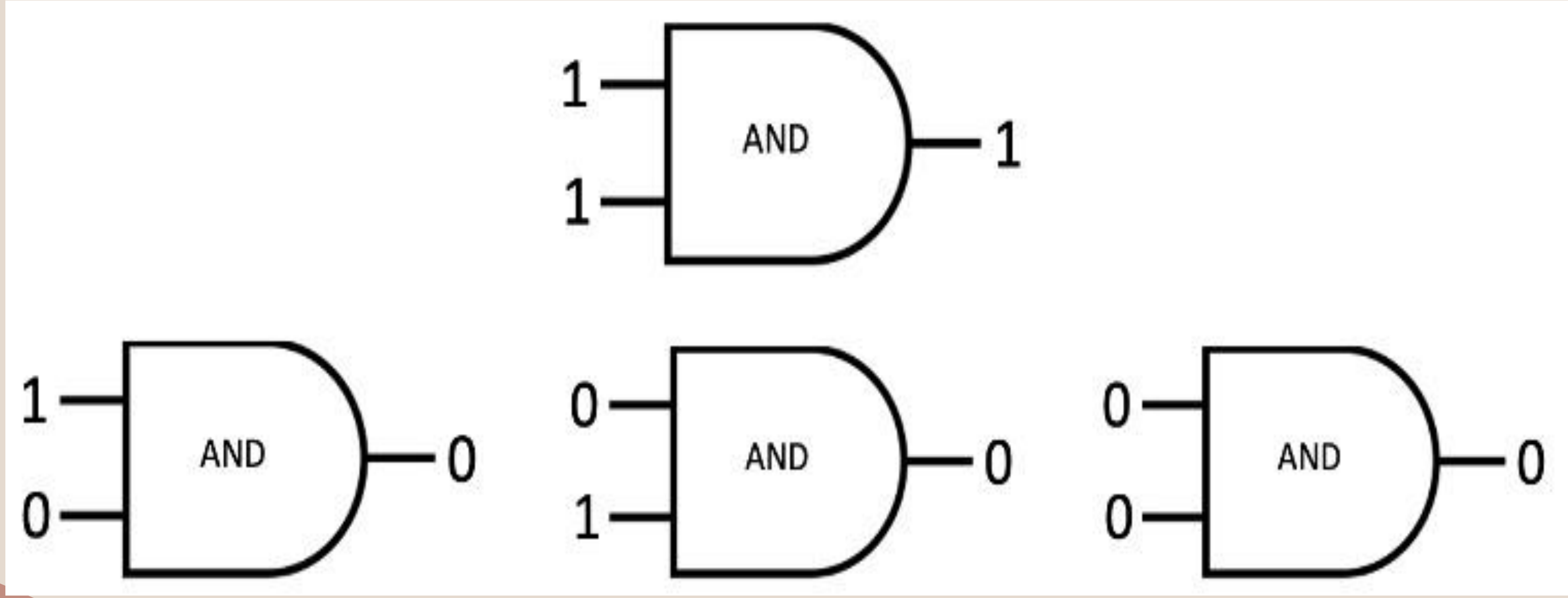
- Quantum applications are built as quantum circuits using quantum gates (like classical computer applications)
- In classical computers, the gates are simple and appear only in architecture & computer organization courses
- As for quantum computing, the gates are more complex and used directly in high level quantum applications (fore & front)
- Due to the complexity of qubits, most of the quantum gates are mathematical and based linear and matrix algebra
- In this lecture we describe samples of well known quantum gates used to build quantum applications

Operator	Gate(s)	Matrix
Pauli-X (X)		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$

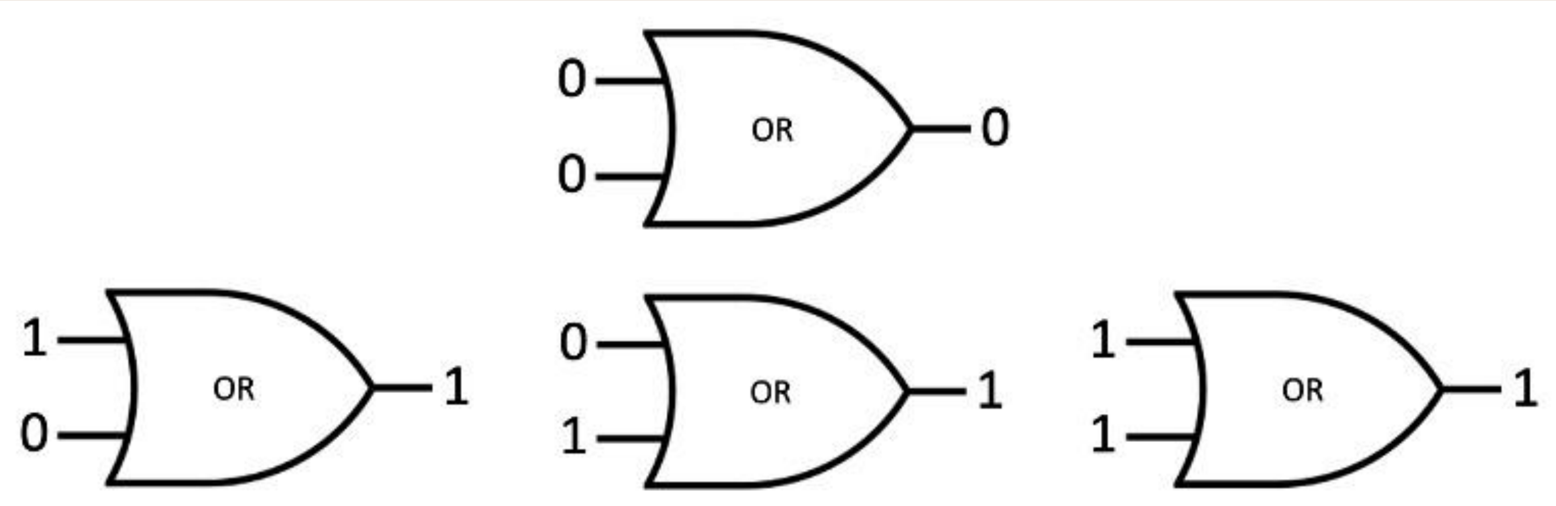
The Classical NOT Gate



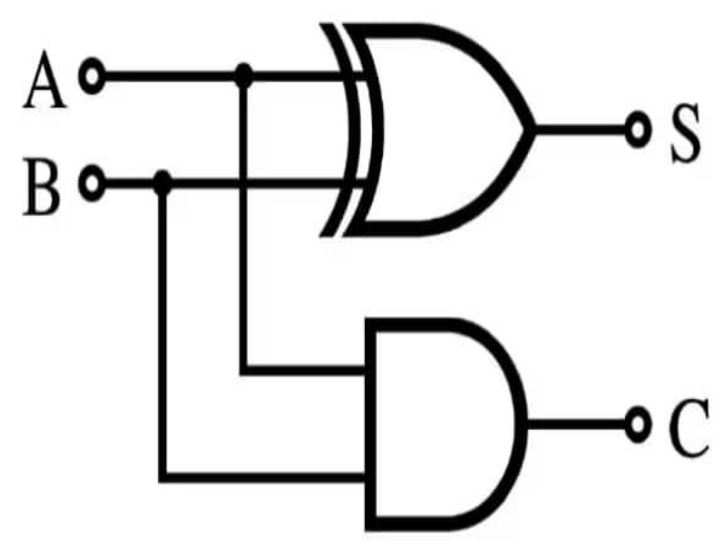
Classical AND gate

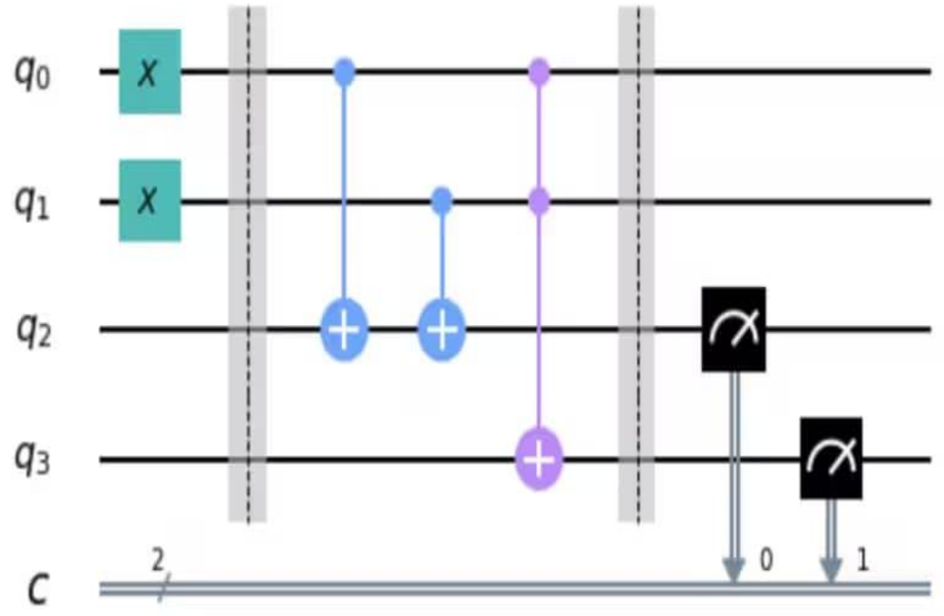


The Classical OR Gate



Two examples of Circuits: Classical VS Quantum





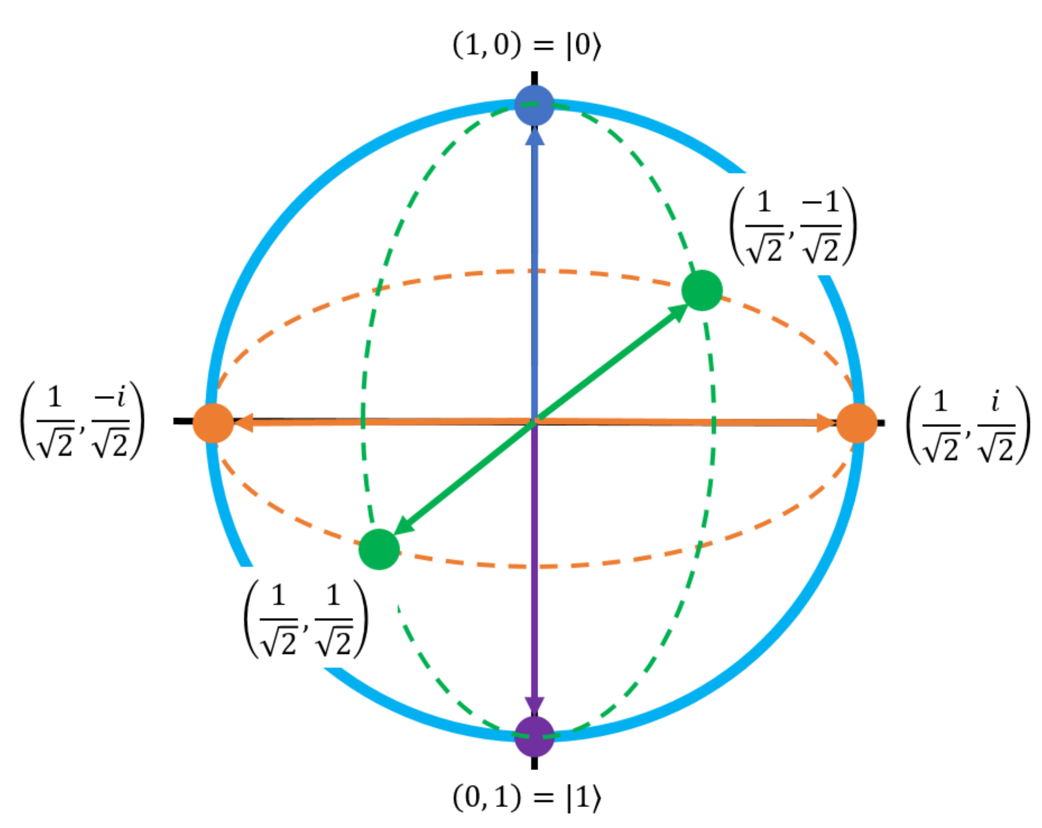
The Bloch Sphere

- This nice simple figure on the right is Bloch Sphere
- The Bloch Sphere is unitary figure used to represent the state of a qubit

The vertical line in the middle is the Z axis (Pole axis)

The green line in the middle is the X axis

The orange arrow is the Y axis



Quantum State and the Bloch Sphere

A qubit's state is represented as a superposition of $|0\rangle$ and $|1\rangle$ .An arbitrary quantum state is represented as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where α and β are complex numbers such that $|\alpha|^2 + |\beta|^2 = 1$.

$|0\rangle$ and $|1\rangle$ are vectors in the two-dimensional complex vector space:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Therefore, an arbitrary quantum state is also represented as

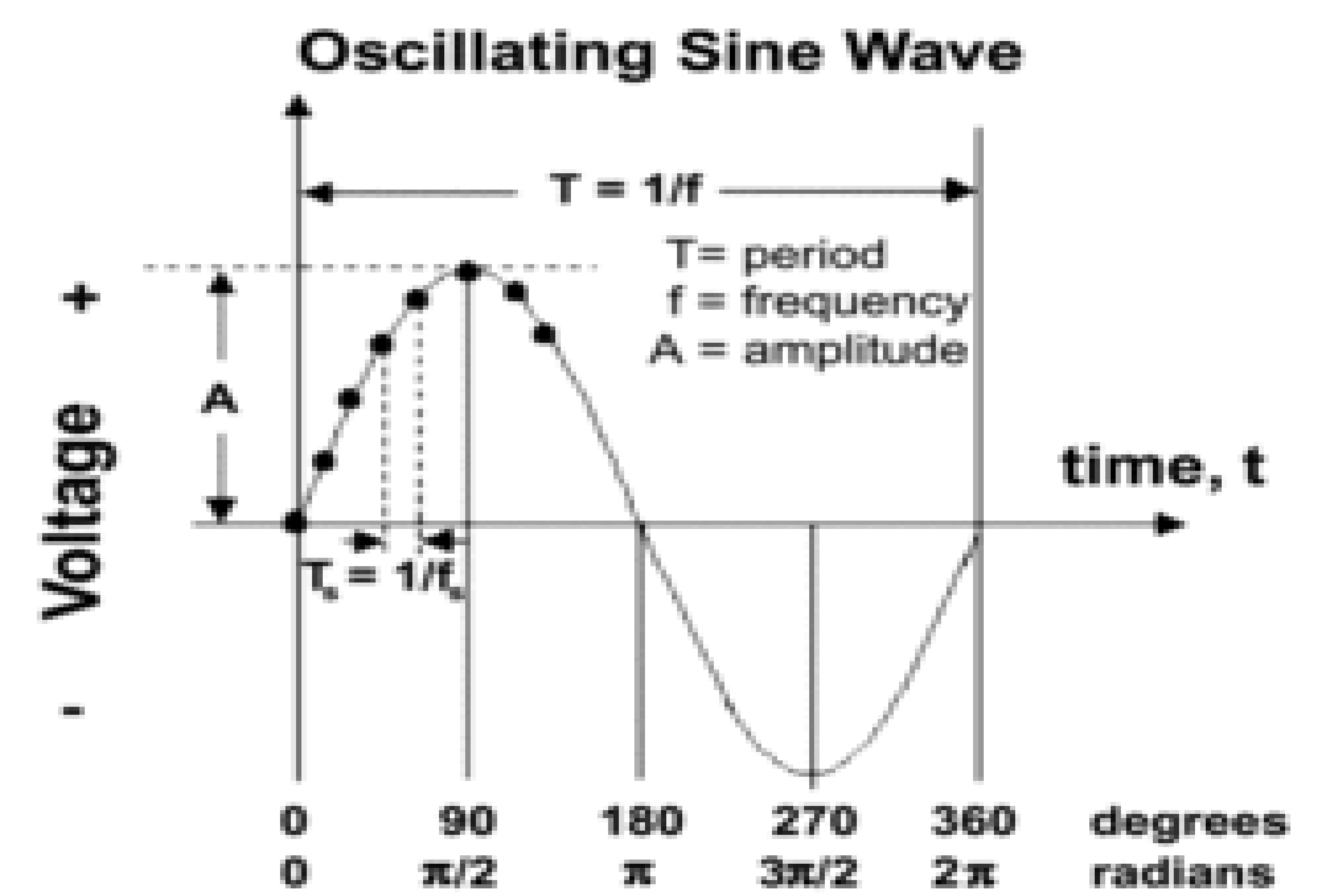
$$|\psi\rangle = \alpha\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$|\psi\rangle=(\alpha\beta)$ is also called the state vector.

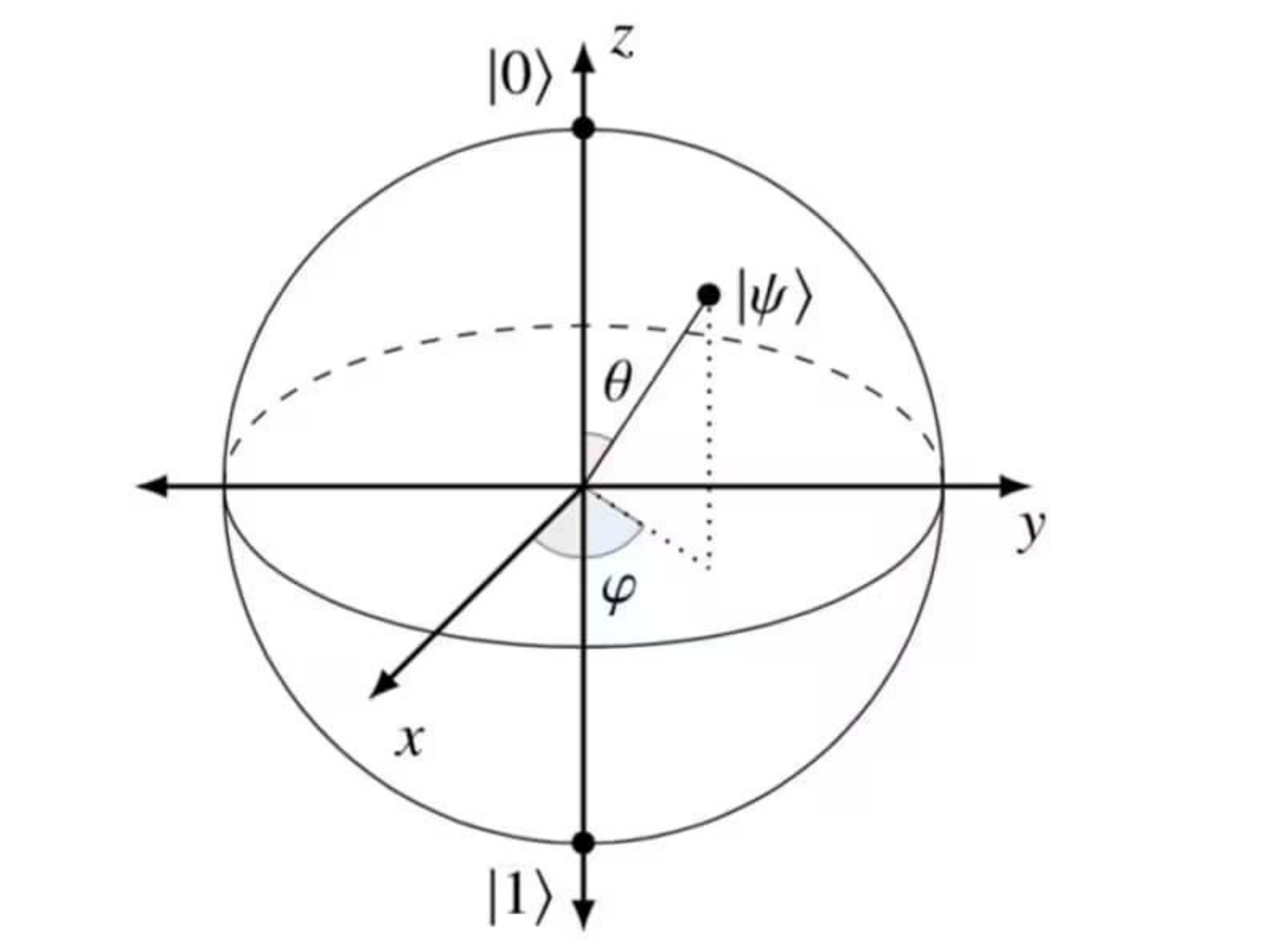
Using radian angles: the Bloch Sphere

- A single-qubit quantum state is also represented as
- $| \psi\rangle = \cos\frac{\theta}{2} | 0\Big\rangle + e^{i\varphi} \sin\frac{\theta}{2} | 1\Big\rangle = \left(\begin{pmatrix} \cos\frac{\theta}{2} \\ e^{i\varphi} \sin\frac{\theta}{2} \end{pmatrix}\right)$
- where θ and φ are the angles of the Bloch sphere in the following figure. (next page)
- Note that Alpha and Beta has been replaced by the radian expressions

Some Concepts Regarding Waves



The Bare Bloch Sphere



The Hadamard Gate

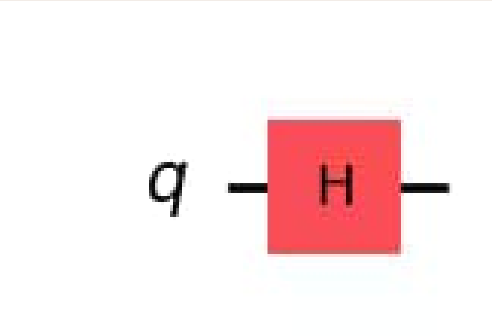
- The Hadamard gate is a π rotation around an axis halfway between the x and z axes on the Bloch sphere.
- Applying the H gate to $|0\rangle$ creates a superposition state such as $\frac{|0\rangle+|1\rangle}{\sqrt{2}}$ .The matrix representation of the H gate is below.

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Apply the Hadamard Gate and Draw Circuit

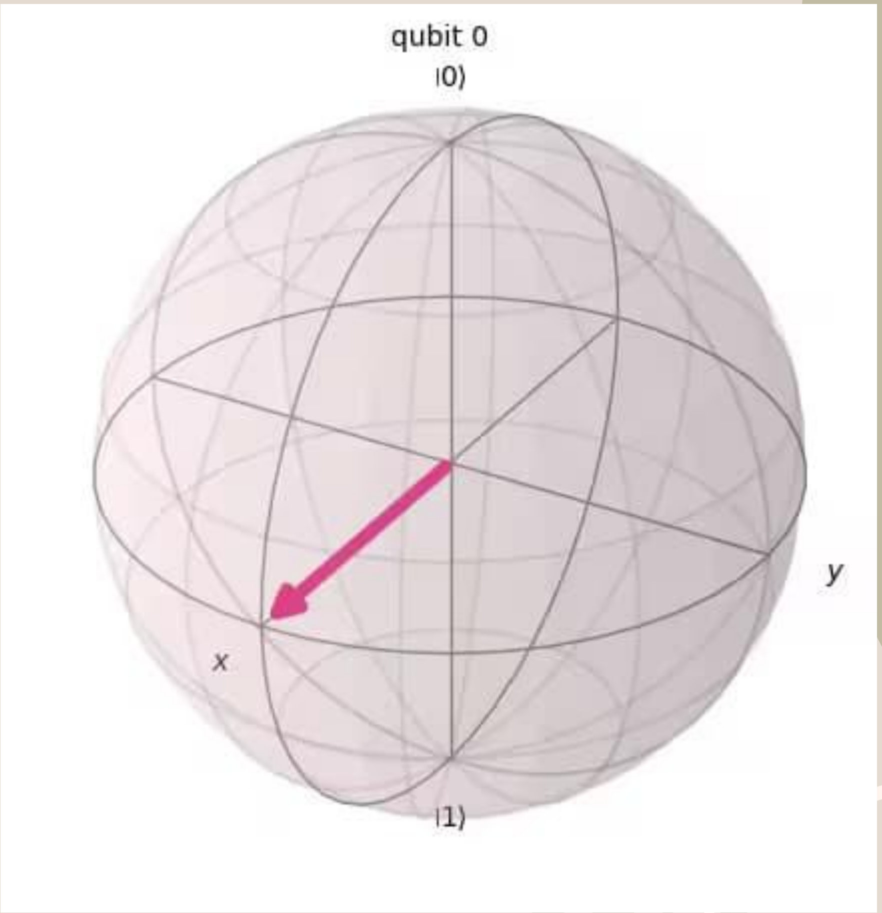
```
qc = QuantumCircuit(1)  # Create the single-qubit quantum
  circuit

# Apply an Hadamard gate to qubit 0
qc.h(0)

# Draw the circuit
qc.draw(output="mpl")
```

- The Circuit:



H on the Bloch Sphere : The $|+\rangle$ state

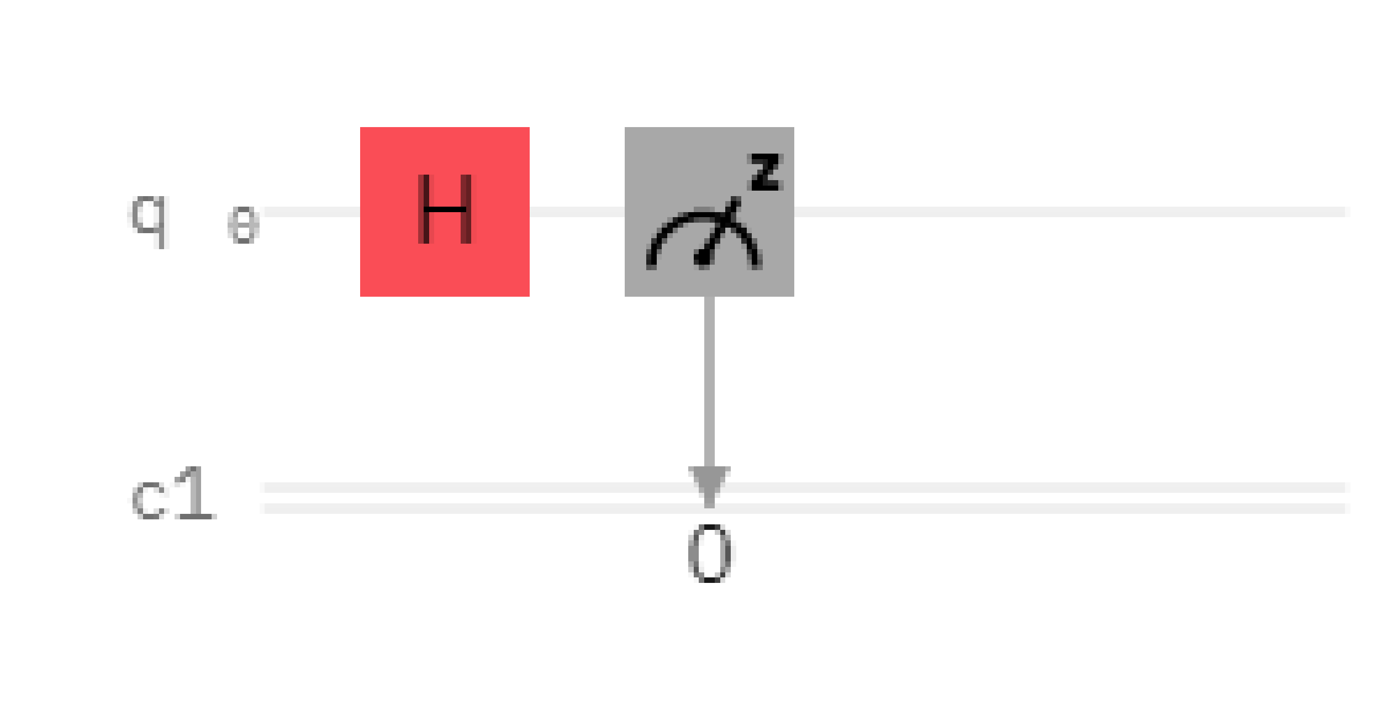
```
# See the statevector
out_vector = Statevector(qc)
print(out_vector)

# Draw a Bloch sphere
plot_bloch_multivector(out_vector)
```



The $|+\rangle \equiv \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

```
Statevector([0.70710678+0.j, 0.70710678+0.j],
            dims=(2,))
```

The Hadamard Gate Circuit



The Hamdard Gate in Action

```
from qiskit import QuantumCircuit, transpile, ClassicalRegister, QuantumRegister
from qiskit_aer import AerSimulator
backend = AerSimulator() # Using local Aer simulator

q = QuantumRegister(1,'q') # Initialise quantum register
c = ClassicalRegister(1,'c') # Initialise classical register
circuit = QuantumCircuit(q,c) # Initialise circuit
circuit.h(q[0]) # Put Qubit 0 in to superposition using hadamard gate
circuit.measure(q,c) # Measure qubit
circ = transpile(circuit, backend) # Rewrites the circuit to match the backend's
  basis gates and coupling map
shots = 1024    # Run and get counts

result = backend.run(circ,shots=shots).result()
counts = result.get_counts(circ)
print('RESULT: ',counts) # Print result
```

The H Gate

Computing $H\mid 0\rangle$

$$H\mid 0\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}0.707\\0.707\end{pmatrix} = \frac{1}{\sqrt{2}}(\mid 0\rangle + \mid 1\rangle)$$

- This superposition state is so common and important, that it is given its own symbol:

$$\mid +\rangle \equiv \frac{1}{\sqrt{2}}(\mid 0\rangle + \mid 1\rangle).$$

- By applying the H gate on the $\mid 0\rangle$ ‹we created a superposition of $\mid 0\rangle$ and $\mid 1\rangle$ where measurement in the computational basis (along z in the Bloch sphere picture) would give you each state with equal probabilities.

The $\mid -\rangle$ state

You might have guessed that there is a corresponding $\mid -\rangle$ state:

$$\mid -\rangle \equiv \frac{\mid 0\rangle - \mid 1\rangle}{\sqrt{2}}.$$

H on the Bloch Sphere: The $|-\rangle$ state

- To get the $|-\rangle$ state on the Bloch Sphere apply the Hadamard gate to the $|1\rangle$ state

The X Gate (NOT Gate)

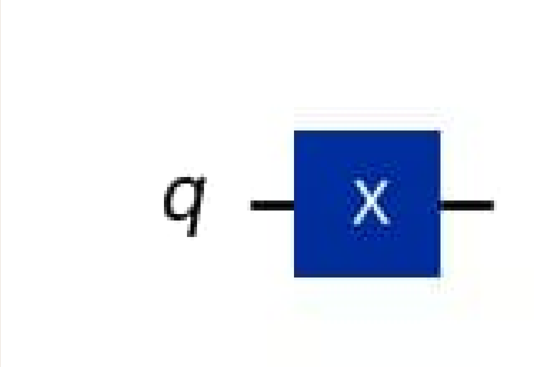
- The X gate is a π rotation around the x axis of the Bloch sphere.
- Applying the X gate to $|0\rangle$ results in $|1\rangle$ ‹and applying the X gate to $|1\rangle$ results in $|0\rangle$ ‹so it is an operation similar to the classical NOT gate, and is also known as bit flip.
- The matrix representation of the X gate is below.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

The X Gate Python Implementation

```
qc = QuantumCircuit(1)  # Prepare the single-qubit quantum
 circuit

# Apply  X gate to qubit 0
qc.x(0)

# Draw the circuit
qc.draw("mpl")
```

The X circuit diagram



The X circuit operator

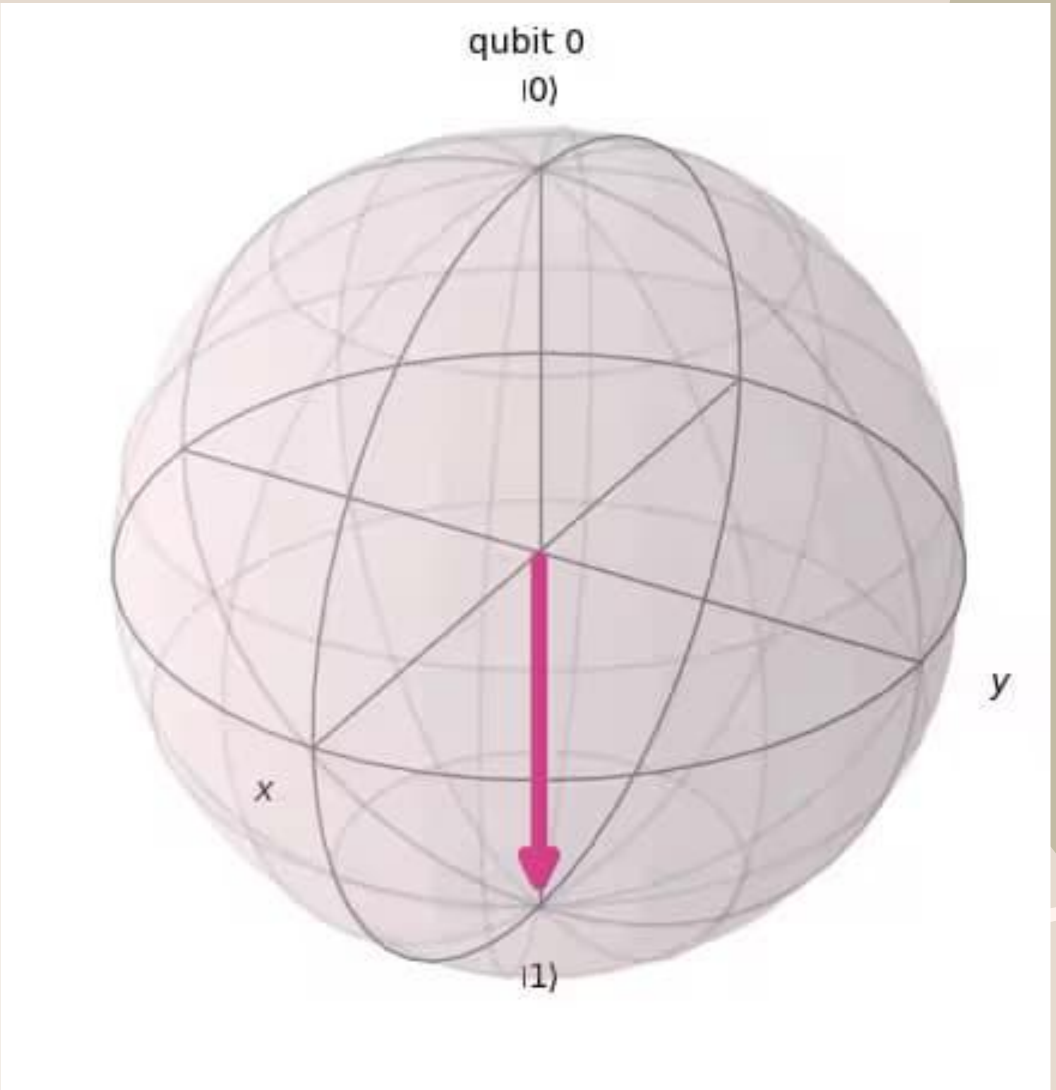
- the initial state is set to $| 0\rangle$ ‹so the quantum circuit above in matrix representation is

$$X | 0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} =| 1\Bigg\rangle$$

Drawing the Bloch Diagram

```
# See the statevector
out_vector = Statevector(qc)
print(out_vector)
 # Draw a Bloch sphere
plot_bloch_multivector(out_vector)
```

```
Statevector([0.+0.j, 1.+0.j],
            dims=(2,))
```



Pauli-Y Gate

Similarly to Pauli-X gate, the Y gate represents a rotation of around the y axis by π radians.

It transforms $|0\rangle$ to $i|1\rangle$ and $|1\rangle$ to $-i|0\rangle$.

$$-\boxed{Y}- = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = i\,|1\rangle\langle 0| - i\,|0\rangle\langle 1|$$

Pauli-Z Gate

The Z gate is actually a special case of the phase shift gate where $\phi = \pi = 180°$.

It represents a rotation around the z axis by π radians.

It has no effect on $|0\rangle$ but transforms $|1\rangle$ to $-|1\rangle$.

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

I-Gate

This is simply a gate that does nothing. Its matrix is the identity matrix.

circuit should have no effect on the qubit state.

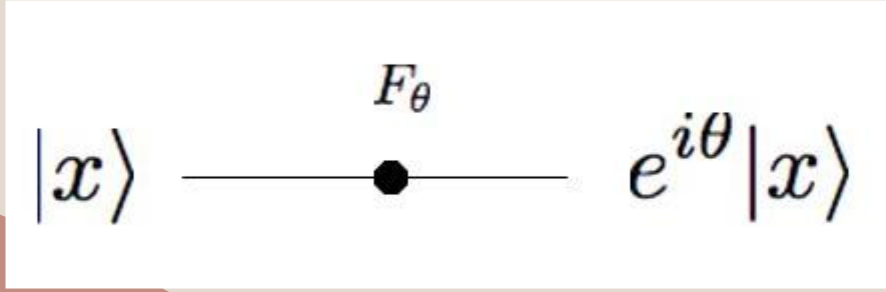
It is often used in calculations, for example: proving the X-gate is its own inverse:

It is considering real hardware to specify a 'do-nothing' or 'none' operation.

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Phase Shifter Gate

- This is a family of single-qubit gates that leave the basis state |0> unchanged and map |1> to e^iϕ|1>.
- The probability of measuring a|0> or |1> is unchanged after applying this gate, however it modifies the phase of the quantum state.
- This is equivalent to tracing a horizontal circle (a line of latitude) on the Bloch sphere by ϕ radians.



$$F_\theta = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix}$$

S and S† Gates

The Phase gate (S gate) is a single-qubit operation.

The S gate is also known as the phase gate or the Z90 gate, because it represents a 90-degree rotation around the z-axis.

The conjugate transpose of S gate is *S*† gate.

$$S = \begin{bmatrix} 1 & 0 \\ 0 & e^{\frac{i\pi}{2}} \end{bmatrix}, \quad S^\dagger = \begin{bmatrix} 1 & 0 \\ 0 & e^{-\frac{i\pi}{2}} \end{bmatrix}$$

T and T† Gates

The T-gate is a very commonly used gate, it is an R_ϕ-gate with $\phi = \pi/4$.

As with the S-gate, the T-gate is sometimes also known as (Z)^(1/4) gate.

The conjugate transpose of T gate is *T*† gate.

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{\frac{i\pi}{4}} \end{bmatrix}, \quad T^\dagger = \begin{bmatrix} 1 & 0 \\ 0 & e^{-\frac{i\pi}{4}} \end{bmatrix}$$

General U gates

- Z, S & T-gates were all special cases of the more general R_ϕ-gate.
- The U_3-gate is the most general of all single-qubit quantum gates.
- It is a parameterized gate of the form:

$$U_3(\theta, \phi, \lambda) = \begin{bmatrix} \cos(\theta/2) & -e^{i\lambda}\sin(\theta/2) \\ e^{i\phi}\sin(\theta/2) & e^{i\lambda+i\phi}\cos(\theta/2) \end{bmatrix}$$

$$U_3(\tfrac{\pi}{2}, \phi, \lambda) = U_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -e^{i\lambda} \\ e^{i\phi} & e^{i\lambda+i\phi} \end{bmatrix} \qquad U_3(0, 0, \lambda) = U_1 = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\lambda} \end{bmatrix}$$

Multiple Qubits Quantum Gates

Swap Gate

Controlled Gates

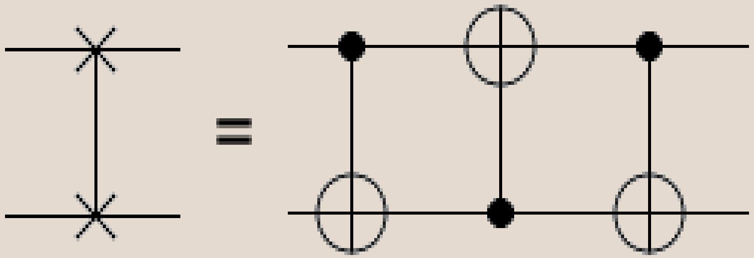
Toffoli (CCNOT) Gate

Fredkin (CSWAP) gate

SWAP Gate

- The swap gate swaps two qubits.
- With respect to the basis |00> |01>, |10>, |11>.
- It is represented by the matrix:

$$\mathrm{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



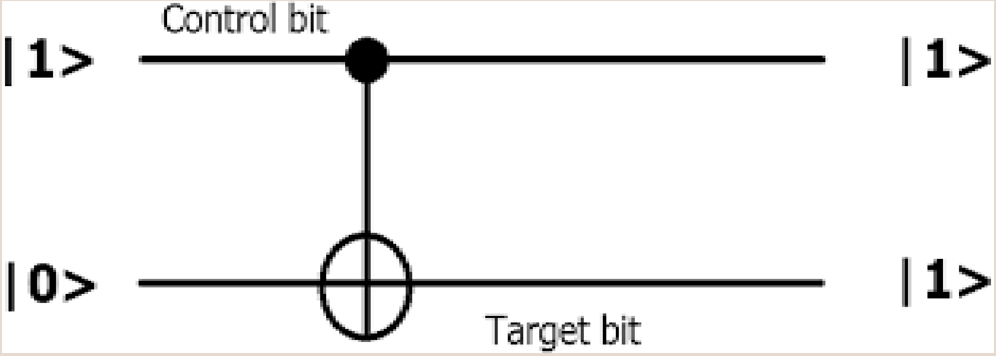
Controlled Gates

Controlled gates require at least one control and one target qubit.

The gate in question will only operate on the target qubit if the control qubit is in a specific state.

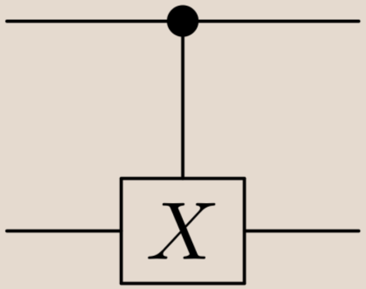
Types of control gates:

- Cx Gate
- Cy Gate
- Cz Gate



Cx Gate

- Cx leaves the control qubit unchanged and performs a Pauli-X gate on the target qubit when the control qubit is in state $|1\rangle$, leaves the target qubit unchanged when the control qubit is in state $|0\rangle$.

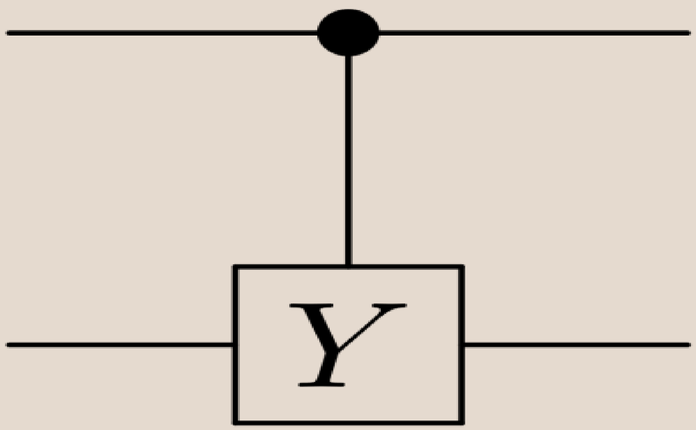


$$CX\begin{bmatrix}1\\0\\1\\0\end{bmatrix}=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{bmatrix}\begin{bmatrix}1\\0\\1\\0\end{bmatrix}=\begin{bmatrix}1\\0\\0\\1\end{bmatrix}$$

Cy Gate

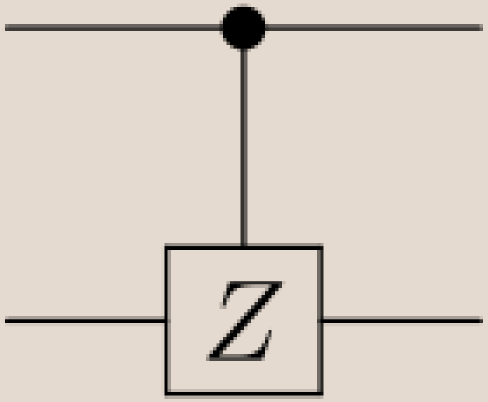
The controlled or Cy gate is another well-used two-qubit gate.

Just as the CNOT applies an Y to its target qubit whenever its control is in state $|1\rangle$, the controlled-Y applies a Y in the same case.



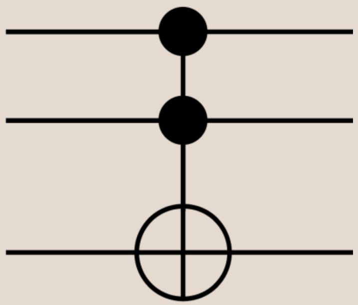
Cz Gate

- The controlled-Z or Cz gate is another well-used two-qubit gate.
- Just as the CNOT applies an X to its target qubit whenever its control is in state $|1\rangle$, the controlled-Z applies a Z in the same case.



Toffoli Gate

- The Toffoli gate is a three qubit gate with two controls and one target.
- It performs an X on the target only if both controls are in the state $|1\rangle$.
- The final state of the target is then equal to either the AND or the NAND of the two controls, depending on whether the initial state of the target was $|0\rangle$ or $|1\rangle$.
- A Toffoli can also be thought of as a controlled-controlled-NOT, and is also called the CCX gate.



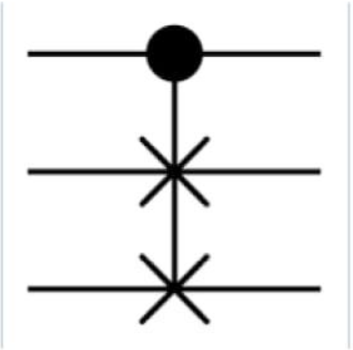
$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Fredkin Gate

It is controlled-SWAP gate.

It only swaps the values of the second and third bits only if the first bit is set to 1.

It is also a reversible gate.



$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

A tool for visualization of Bloch Sphere

- Please find a nice tool for visualizing and understanding Bloch Sphere representations

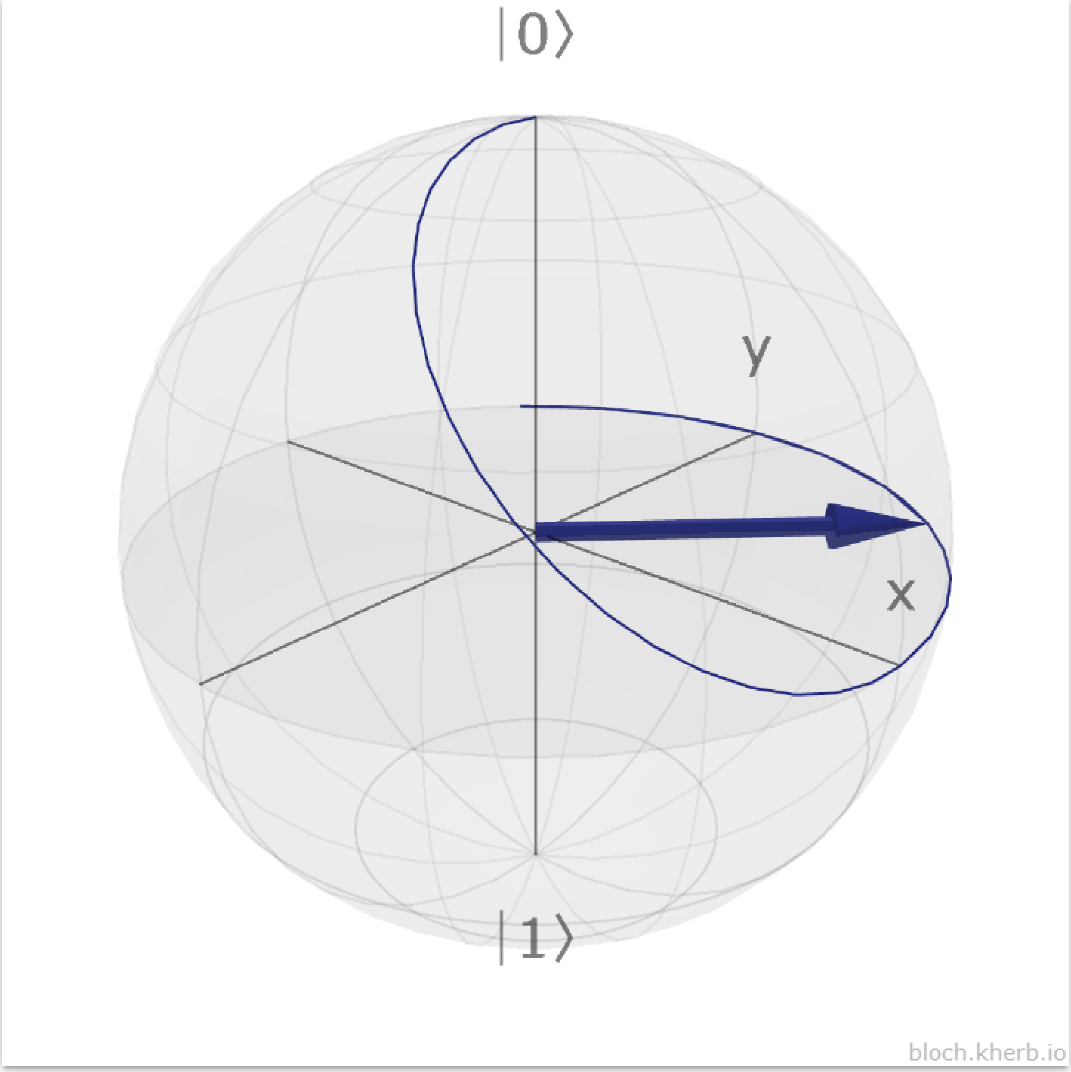
Bloch sphere visualizer

https://bloch.kherb.io

Bloch sphere simulator

Tool Explanation GitHub ↗

INIT UNDO



DOWNLOAD

Rotations around default axes

Rotations around custom axis

Quantum gates

Pauli gates	X Y Z
Hadamard	H
Phase	S S† T T†

Pulses

Settings